COURSE INFORMATION

Department/ Faculty:	Software Engineering/Computing	**Page:**	1 of 6
Course code:	SECJ2154	**Academic Session/Semester:**	2024/2025-2
Course name:	Object Oriented Programming	**Pre/co requisite (course name and code, if applicable):**	Programming Technique 1 Programming Technique 2
Credit hours:	4		

Course synopsis	This course presents the concepts of object orientation and object-oriented programming techniques using Java programming language. It provides students with a thorough look at the basic constructs of the Java programming language such as its basic data types and operations. It also emphasizes on the use of standard Java APIs that allow students to develop text-based and GUI applications. It will also provide the programming techniques on exception handling and input/output files. At the end of this course, students should be able to use the basic constructs in object-oriented programming and utilize the selected Java APIs			
Course coordinator	Zuraini Ali Shah			
Course lecturer(s)	**Name**	**Section(s)**	**Room No.**	**E-mail**
	Dr. Norsham Idris	01	N28-Level 3	norsham@utm.my
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	Dr. Muhammad Khatibsyarbini	03		khatibsyarbini@utm.my
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	Ms. Lizawati binti Mi Yusuf	05	N28-Level 4	lizawati@utm.my
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	PM Dr Shahida Sulaiman	10		shahidasulaiman@utm.my
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	Dr. Fazliaty Edora binti Fadzli	12	N28-Level 2	fazliaty.edora@utm.my

Prepared by: Name: Zuraini Ali Shah (Course Coordinator) Signature: Date: 6 March 2025	Certified by: Name: PM. Dr. Radziah Mohamed (Director of Software Engineering) Signature: Date:

Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching & Learning (T&L) methods and Assessment methods:

No.	Course Learning Outcome	*Program Learning Outcome	Weight (%)	**Taxonomies and generic skills	T&L methods	***Assessment methods
CLO 1	Apply OOP concepts in problem-solving and develop Java applications.	PO1 (Knowledge Understanding)	55	C3	L, T	T1, F, Q
CLO 2	Develop Java applications that leverage the object-oriented features of the Java language such as encapsulation, inheritance and polymorphism in a given time frame.	PO3 (Problem Solving)	35	P4	L, P	PR, T2, LE
CLO 3	Work in a team to develop a medium to complex program as a group mini project using Java programming language.	PO5 (Thinking Skill)	10	P4	P	PR, Pre

*Program Learning Outcome
PO1 - Ability to acquire and apply theory and principles of computer science and equip with social science and personal development knowledge.
PO3 - Ability to design and construct computer programs using standard approaches.
PO5 - Ability to work effectively in a team

**Taxonomies and Generic Skills
*** LE – Lab exercise, Q – Quiz, A – Assignment, T1 – Test 1, T2-Test 2, F – Final Exam, PR – Mini Project, Pre – Presentation

L – Lecture, T – Tutorial, P – Practical

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class activities
2.	Independent Study	E-learning
3.	Project-based learning	Conducted through study assignments and mini projects. Tasks are given in sequential steps throughout the semester. Students in a group of 3 are given a case study that requires to use of object-oriented programming concepts to solve certain problems efficiently.

Weekly Schedule:

Week 1 17 - 21 March	**Chapter 1 Introduction to Object-oriented concepts and a general overview of Java API** ● Object oriented definition ● Object oriented concept Hari Keputeraan Johor (23 March – Sunday)
Week 2 24 – 29 March	**Chapter 1 (continued)** ● Classes in Java ● Java Basics ● Creating Java Programs ● Java Variables and data types Cuti Umum Hari Keputeraan Johor (24 March – Monday)
Week 3 31 March - 4 April	**Chapter 2: Introduction to Classes and Objects (QUIZ 1, LE 1)** ● Instance Fields, Accessor and Mutator Methods ● Constructors ● UML class diagram ● Static Class Members ● Passing and returning Objects to/from Methods Hari Raya Aidilfitri (31 March-Monday , 1April -Tuesday)
Week 4 7 April - 11 April	**Chapter 3: Enumerated Type, Wrapper Class and Java Package (LE 2)** ● Enumerated Types ● Garbage Collection ● Wrapper Classes ● Packages and import Statements
Week 5 14 April – 18 April	**Chapter 4: Arrays** ● Defining One-Dimensional Array ● Passing Arrays as Arguments to Methods ● Returning Arrays from Methods
Week 6 21 – 25 April	**Chapter 4: Arrays (continued) (QUIZ 2, LE 3)** ● String Arrays ● Arrays of Objects ● Defining Two-Dimensional Array
Week 7 28 April – 2 May	**Chapter 5: Vectors & Collections (TEST 1: 28 April, Monday @ 8 - 10 PM, Chapters 1 – 4, N28-BK7)** ● Array Lists ● Vector **Labour Day (1 May – Thursday)**
Week 8 5 May - 10 May	**Mid-Semester Break**

Week 9 12 - 16 May	**Chapter 6: Class Relationships** ● Association ● Aggregation ● Composition Wesak Day (12 May – Monday)
Week 10 19 – 23 May	**Chapter 7: Inheritance (LE 4)** ● Introduction to Inheritance ● Protected Members ● The Object Class
Week 11 26 May – 30 May	**Chapter 7: Inheritance (continued)** ● Calling the Superclass Constructor ● Chains of Inheritance ● Overriding Superclass Methods **(TEST 2: 28 May, Wednesday @ 8 - 10 PM, Chapters 1 – 6, MPK1 - MPK10**
Week 12 2 – 6 June	**Chapter 8: Polymorphism (QUIZ 3)** ● Introduction to Polymorphism ● Abstract Classes and Abstract Methods Hari Keputeraan YDP Agong (2 June – Monday) Hari Raya Haji (7 June – Saturday)
Week 13 9 – 13 June	**Chapter 8: Polymorphism (continued) (LE 5)** ● Interface and implements
Week 14 16 – 20 June	**Chapter 9: Exception Handling** ● Introduction to Exceptions ● Checked vs Unchecked exceptions ● Throwing Exceptions
Week 15 23 – 27 June	**Group Project Demo/Presentations** Awal Muharram (27 June- Friday)
Week 16 30 June - 4 July	**REVISION WEEK**
Weeks 17 - 19 **7** - 27 July	**FINAL EXAM WEEKS**

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

Team working and Presentation

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline	Guided Learning (Face to Face)				Teaching and Learning Activities		TOTAL SLT
					Guided Learning Non-Face to Face	Independent Learning Non-Face to face	
CLO	**L**	**T**	**P**	**O**			
CLO 1	21h	15h	9h		3h	15h	**63h**
CLO 2	11h	9h	13h		3h	10h	**46h**
CLO 3	2h		8h		3h	5h	**18h**
Total SLT	**34h**	**24h**	**30h**		**9h**	**30h**	**127h**

No	Continuous Assessment	PLO	Percentage	Total SLT
1	Lab Exercises (5)	3	15	**As in CLO2 5.5h**
2	Quiz (3)	1	10	**As in CLO1 1.5h**
3	Test 1	3	15	**As in CLO1 2h**
4	Test 2 (practical/lab)	3	15	**As in CLO2 2h**
5	Mini Project	3	5	**As in CLO2 9h**
6	Mini Project	5	5	**As in CLO3 8h**
7	Presentation	5	5	**As in CLO3 2h**
	Final Assessment	**PLO**	**Percentage**	**Total SLT**
1	Final Exam (Written)	1	30	**As in CLO1 3h**

Grand Total SLT	**160h**

L: Lecture, T: Tutorial, P: Practical, O: Others

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

1. Software: JAVA SDK (latest version) - https://www.oracle.com/my/java/technologies/downloads/
2. Editor: https://code.visualstudio.com/download
3. UML Diagram Tool: https://app.diagrams.net/; https://plantuml.com/

Learning Resources:

Main references:

Norazah Yusof, Radziah Mohamad, dan Nor Bahiah Hj. Ahmad. Object Oriented Programming Using Java.7th Edition. 2014. Penerbit UTM.

Additional references:

1. Y. Daniel Liang, Introduction to Java programming. 8th edition. 2011. Pearson - Prentice Hall.
2. Deitel, Deitel, Java How to Program, 7th edition. 2007. Pearson – Prentice Hall.
3. Tony Gaddis, Starting out with Java – From Control Structures through Objects, 3rd edition. 2008. Addison Wesley.
4. Walter Savitch, Absolute Java, 2nd edition. 2006. Pearson Addison-Wesley.
5. Joyce Farrel, Java Programming. 4th edition. 2008. Thomson Course Technology.

Academic honesty and plagiarism:

Copying of work (texts, lab results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and exams and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Student with <u>less than 80%</u> of total attendance is not allowed to sit for final exam.
2. Students are required to behave and follow the University's dressing regulation and etiquette all the time.
3. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted **<u>three days</u>** after the due date will not be accepted.

Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given <u>within one week</u> of the initial date of exam.

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